

Amel Derras-Chouk

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EDUCATION

- 2016–2022 PhD in Physics at CUNY Graduate Center, New York, NY
Dissertation: *Collapse of Two-Dimensional Magnetic Skyrmions*
Advisors: Dist. Prof. Eugene M. Chudnovsky and Prof. Dmitry A. Garanin
- 2011–2015 B.S. in Physics and Mechanical Engineering at Tufts University

RESEARCH EXPERIENCE

- Postdoctoral Research Fellow**, *Columbia Climate School* Sept. 2024–present
- Develop physical models to explain emergent ordered behavior in tropical anvil clouds
 - Process remote sensing data to validate analytical models with cloud observations
- Postdoctoral Researcher**, *The City College of New York* Sept. 2022–Aug. 2024
- Combined remote sensing and physical models to quantify cumulus convection, estimating uncertain atmospheric variables like vertical velocity and mass flux
- Graduate Student Researcher**, *NASA GISS* Oct. 2021–Aug. 2022
- Automated retrieval and analysis of over 40 years of Landsat optical data using Python
 - Proposed a methodology based on literature to measure flooding using Sentinel-1 data
 - Integrated data from NOAA, USGS, ASF, and citizen scientists into a cohesive workflow
 - Tailored presentations to heterogeneous audiences of experts and local teachers
- Graduate Student Researcher**, *CUNY Graduate Center* Jun. 2017–Aug. 2022
- Developed classical and quantum theories to model two-dimensional spin systems
 - Parallelized thousands of computations using multithreading and distributed computing
 - Analyzed program outputs by visualizing results and performing statistical tests
 - Presented results to audiences of scientific backgrounds at national conferences

MANUSCRIPTS SUBMITTED AND IN PREPARATION

- Derras-Chouk, Amel, Zhengzhao Johnny Luo, and Gregory Elsaesser (In prep). *A Satellite-based Estimate of Deep Convective Precipitation Efficiency in the Tropics*.
- Derras-Chouk, Amel et al. (Submitted). *Estimating Vertical Velocity from Temperature, Pressure, and Latent Heating*.

PUBLICATIONS

- Braneon, Christian et al. (2024). “NPCC4: New York City climate risk information 2022—observations and projections”. In: *Annals of the New York Academy of Sciences* 1539.1, pp. 13–48.

- Derras-Chouk, Amel and Zhengzhao Johnny Luo (2024). “A geostationary satellite-based approach to estimate convective mass flux and revisit the hot tower hypothesis”. In: *Surveys in Geophysics* 45.6, pp. 1959–1977.
- Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2022). “Dynamics of the collapse of a ferromagnetic skyrmion in a centrosymmetric lattice”. In: *Physical Review B* 105.13, p. 134432.
- Derras-Chouk, Amel and Eugene M Chudnovsky (2021). “Skyrmions near defects”. In: *Journal of Physics: Condensed Matter* 33.19, p. 195802.
- Derras-Chouk, Amel, Eugene M Chudnovsky, and Dmitry A Garanin (2021). “Quantum states of a skyrmion in a two-dimensional antiferromagnet”. In: *Physical Review B* 103.22, p. 224423.
- (2019). “Thermal collapse of a skyrmion”. In: *Journal of Applied Physics* 126.8.
 - (2018). “Quantum collapse of a magnetic skyrmion”. In: *Physical Review B* 98.2, p. 024423.
- Derras-Chouk, Amel, Eugene M Chudnovsky, Dmitry A Garanin, and Reem Jaafar (2018). “Graphene cantilever under Casimir force”. In: *Journal of Physics D: Applied Physics* 51.19, p. 195301.

SELECTED PRESENTATIONS

- Derras-Chouk, A., Z. J. Luo, G. Elsaesser, T. Matsui, A. F. Prein, and J. Wu, 2025: Estimating vertical velocity from temperature, pressure, and latent heating. Poster presentation delivered at Gordon Research Conference on Atmospheric Radiation, Lewiston, ME, USA
- Derras-Chouk, A., Z. J. Luo, and G. Elsaesser, 2025: Estimating convective mass flux, detrainment, and precipitation efficiency using CloudSat, Geostationary, and IMERG data. Oral presentation delivered at CloudSat Calipso Science Team Meeting, Fort Collins, CO, USA
- Derras-Chouk, A., and Z. J. Luo, 2023: A Satellite-Based Approach to Estimating Convective Mass Flux and Revisiting the Hot Tower Hypothesis. AGU, San Francisco, CA.

TEACHING EXPERIENCE

- Adjunct Lecturer, Lehman College** Aug. 2017–May. 2021
- Taught physics labs to over 50 students a semester, emphasizing experimental design
 - Assessed student understanding through written assignments and problem-solving sessions
 - Redesigned seven experiments for remote learning and incorporated student feedback

HONORS AND AWARDS

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| Graduate Center Dissertation Fellowship | 2020–2021 |
| Physics Tithe Fellowship | 2018–2022 |